EUCAST EUROPEAN COMMITTEE ON ANTIMICROBIAL SUSCEPTIBILITY TESTING

European Society of Clinical Microbiology and Infectious Diseases

Standard Operating Procedure

MIC distributions and the setting of epidemiological cut-off (ECOFF) values.

EUCAST SOP 10.2

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Foreword

The European Committee on Antimicrobial Susceptibility Testing (EUCAST) is organised by the European Society for Clinical Microbiology and Infectious Diseases (ESCMID), the European Centre for Disease Prevention and Control (ECDC), and the active national antimicrobial breakpoint committees in Europe. EUCAST was established by ESCMID in 1997, was restructured in 2001-2002 and has been in operation in its current form since 2002. Countries outside Europe are increasingly forming EUCAST-affiliated national antimicrobial susceptibility testing (AST) committees and implementing the EUCAST systems of testing methods and breakpoints.
The current remit of EUCAST is to harmonise clinical breakpoints for existing antimicrobial agents in Europe, to determine clinical breakpoints for new agents, to set epidemiological (microbiological) breakpoints, to revise breakpoints as required, to harmonise methodology for antimicrobial susceptibility testing, to develop a website with MIC and zone diameter distributions of antimicrobial agents for a wide range of organisms, to provide continuing education on AST and to liaise with European governmental agencies and European networks involved with antimicrobial resistance and resistance surveillance.

Information on EUCAST, EUCAST breakpoints and all documents are freely available on the EUCAST website at http://www.EUCAST.org.

Citation of EUCAST documents

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EUCAST documents published on the EUCAST website should be cited in the following way: European Committee on Antimicrobial Susceptibility Testing. Name of document, EUCAST version number, year. Website address.

This SOP should be cited as: “European Committee on Antimicrobial Susceptibility Testing. MIC distributions and epidemiological cut-off value (ECOFF) setting, EUCAST SOP 10.2, 2021. http://www.eucast.org.

Abbreviations

EUCAST	European Committee on Antimicrobial Susceptibility Testing
ECOFF	Epidemiological cut-off value
TECOFF	Tentative epidemiological cut-off value
MIC	Minimum inhibitory concentration
ISO	International Organization for Standardization
MHF	Mueller-Hinton F (broth or agar)
MRSA	Methicillin-resistant *Staphylococcus aureus*
MSSA	Methicillin-susceptible *Staphylococcus aureus*
ESBL	Extended-spectrum beta-lactamase

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1	Scope
1.1	This SOP describes how EUCAST manages data on MIC distributions and sets epidemiological cut-off values.
1.2	MIC distribution data are generated through many different processes using a variety of methods. These data are collected by EUCAST for potential aggregation and display on the EUCAST MIC distribution website (https://mic.eucast.org/).
1.3	Distribution data that satisfy specific criteria for aggregation, described in this SOP, are used for the setting of epidemiological cut-off values, which are also published on the website.

2 Introduction

2.1

In 2002, EUCAST introduced the concept of gathering large numbers of MIC values to present as aggregated MIC distributions for each combination of species and antimicrobial agent, to define phenotypic wild type distributions on which to determine the most suitable MIC value to represent the upper end of the wild type distribution, defined as the epidemiological cut-off value.

The distributions are for bacteria and fungi and for antimicrobial agents that are or have been used in human and veterinary medicine. The MIC distributions are from a wide variety of sources including breakpoint committees, individual researchers in human and veterinary medicine, programmes for the surveillance of antimicrobial resistance in humans and animals and from EUCAST development projects, or have been put together by pharmaceutical companies during programmes for the development of new agents. There are currently more than 30 000 MIC distributions in the EUCAST database, which amounts to several million MIC values.

Although all distributions are imported to the database, only MIC distributions that adhere to a set of basic conditions have been accepted for aggregation with other distributions (see introductory notes on the EUCAST MIC distribution website (http://mic.eucast.org/Eucast2/), also shown in Appendix 1). An aggregated distribution may consist of data from up to one hundred different sources and of as many as 90000 MIC values (e.g. the aggregated distributions of cefuroxime MICs for *Escherichia coli* and of vancomycin MICs for *Staphylococcus aureus*). The distributions are presented as tables and as histograms (see example in Appendix 2).

For a given microbial species and antimicrobial agent, the epidemiological cut-off value (ECOFF) is *the highest MIC for organisms devoid of phenotypically detectable, acquired resistance mechanisms*. It defines the upper end of the wild-type MIC distribution and is typically written as X mg/L, while the wild type is written as ≤ X mg/L and the non-wild type as > X mg/L. The ECOFF provides EUCAST with a means to minimise the risk of setting clinical breakpoints which split wild-type MIC distributions of organisms. ECOFFs also provide an opportunity to compare rates of acquired resistance with greater precision than with clinical breakpoints and in situations where clinical breakpoints differ (e.g. between organisations, between humans and animals), change over time or have not been set.

There is worldwide keen interest in the EUCAST MIC distribution database and ECOFFs. The EUCAST ECOFFs have been recognised by the World Health Organization (WHO), the World Organisation for Animal Health (OIE), The European Food Safety Authority (EFSA) and the Codex Alimentarius Commission as a very important component of international surveillance of antimicrobial resistance.

It is recognised that the rules for including and excluding MIC distributions and the methods used to set ECOFFs need to be defined and agreed at an international level.

2.2	Inhibition zone diameter distributions and zone diameter ECOFFs are also available on the EUCAST 'Antimicrobial wild-type distributions of microorganisms website' (http://mic.eucast.org/Eucast2/). These are not considered in this SOP, although much of what is stated below concerning MIC distributions and ECOFFs would apply also to inhibition zone diameters.

3	**Standards for data required for setting ECOFFs**
3.1	**Underlying assumptions** 3.1.1 MIC measurement has inherent variability which can be controlled but not eliminated. This becomes evident when an isolate is tested repeatedly under the exact same conditions. Variability increases with the number of laboratories and staff involved, different manufacturers of material used, etc. Variation in replicate testing within a single laboratory is referred to as intra-laboratory variation and variation in replicate testing between laboratories is referred to as inter-laboratory variation. 3.1.2 Both intra- and inter-laboratory variation result from technical variation, including differences between how different individuals perform the same test, how results are read, differences in cell density in the inoculum, differences in how materials (broth microdilution plates, agar plates, etc.) are made or handled, differences between products from different manufacturers, differences in temperature stability of incubators, differences in atmospheres in incubators, etc. 3.1.3 Among isolates without phenotypically-expressed resistance affecting the agent in question, observed variation comprises methodological and biological variation in different proportions depending on the species-antimicrobial agent combination (Mouton et al., *J Antimicrob Chemother* 2018; 73: 2374-9). 3.1.4 Wild-type MIC distributions follow a log-normal distribution which is generally 3-5 two-fold dilutions wide.
3.2	**Acceptance criteria for MIC distributions** 3.2.1 MIC distributions submitted for potential use in setting ECOFFs are individually reviewed against the following criteria for acceptability. Organisms may be from human sources, animal sources, antimicrobial resistance surveillance studies, studies undertaken during the development of new agents, data from food safety projects and data from environmental studies. Data may be from published or unpublished sources as long as methodology and the materials are properly characterized. Data may be from projects where datamining of publications targeting one species or one class of agents has uncovered useful distributions (add origin of data). Isolates may include selected wild-type and resistant organisms from studies of antimicrobial activity. If isolates are divided into those with and without a defined resistance mechanism (e.g. MRSA and MSSA, ESBL producing and non-producing *Enterobacterales*, *Enterococcus* spp. with or without *van* genes) the characteristics of the organisms in each MIC distribution should be clearly defined.

The methods used to determine MICs shall, wherever possible, be the International Organization for Standardization (ISO) broth microdilution reference methods:

> Antibacterial agents: Clinical Laboratory Testing and In Vitro Diagnostic Test Systems. Susceptibility Testing of Infectious Agents and Evaluation of Performance of Antimicrobial Susceptibility Test Devices. Part 1: Reference Method for Testing the In Vitro Activity of Antimicrobial Agents Against Rapidly Growing Aerobic Bacteria Involved in Infectious Diseases. International Standard 20776-1. Geneva, Switzerland: ISO; 2019
>
> Antifungal agents (for yeasts): Clinical laboratory testing and in vitro diagnostic test systems -- Reference method for testing the in vitro activity of antimicrobial agents against yeast fungi involved in infectious diseases. International Standard 16256. Geneva, Switzerland: ISO; 2012. The EUCAST-specified variant to be used

Other acceptable methods include:

> Antifungal agents (for moulds): EDef 9.3.1 (Method for the determination of broth dilution minimum inhibitory concentrations of antifungal agents for conidia forming moulds, January 2017) for conidia-forming moulds. (https://www.eucast.org/astoffungi/methodsinantifungalsusceptibilitytesting/ast_of_moulds/)
>
> Antifungal agents (for dermatophytes): EDef 11.0 (Method for the determination of broth dilution minimum inhibitory concentrations of antifungal agents for conidia forming moulds, January 2017) for conidia-forming moulds. (https://www.clinicalmicrobiologyandinfection.com/article/S1198-743X(20)30526-7/fulltext?dgcid=raven_jbs_aip_emai)
>
> Antimycobacterial agents: EUCAST reference method for MIC determination of Mycobacterium tuberculosis (http://www.eucast.org/fileadmin/src/media/PDFs/EUCAST_files/Mycobacteria/Methods_in_AMST/Technical_protocol_AMST_MIC_reference_method_190719.pdf)
>
> or for any of the above, a method giving equivalent results to the ISO standards. Part 2 of ISO 20776 describes how equivalence to the reference method is determined. If the reference method is other than the ISO broth microdilution method, the accepted method will be the reference method defined by EUCAST, e.g. agar dilution for mecillinam and fosfomycin and agar dilution for defined species (such as most anaerobic bacteria), MHF broth or agar for most fastidious organisms, Fastidious Anaerobe Agar for anaerobic bacteria.

3.2.3 For some classes of organisms there is currently no international standard or EUCAST reference method. In such cases MIC distributions, with or without ECOFFs, will be published with the testing method clearly stated.

3.2.4 The MIC values must conform to a two-fold dilution series based on 1 mg/L. e.g. 0.002, 0.004, 0.008, 0.016, 0.03, 0.06, 0.125, 0.25, 0.5, 1, 2, 4, 8, 16, 32, 64, 128, 256 and 512 mg/L (values below 0.125 are displayed in the rounded version here).

3.2.5 Isolates shall be identified to the level of species or, where this is not possible with modern techniques such as MALDI-TOF), species complex (e.g. *Enterobacter cloacae* complex). Data on species "ending with .spp" are accepted and stored by the software, but are no longer displayed or used for setting ECOFFs.

3.2.6 Any MIC distribution approved for aggregation with other distributions is marked "approved" in the database.

3.2.7 Any MIC distribution which does not qualify for aggregation is marked "rejected". The software allows for adding a flag to explain the reason to curators for rejection as listed below:

- Duplicate distributions are not accepted – rejected and inactivated in database.
- Distributions dominated by isolates from outbreaks are not accepted – rejected and inactivated in database.
- A distribution on an uncertain or poorly defined species is marked "rejected - Invalid method"
- A distribution consisting of less than 15 datapoints in the putative wild type is marked "rejected - Few"
- A distribution where truncation is unacceptable and at the **lower** end of the distribution is marked "rejected - TruncL"
- A distribution where truncation is unacceptable and at the **upper** end of the distribution is marked "rejected - TruncU"
- Aberrant mode. The modal MIC deviates from the most common modal MIC by more than one dilution when compared with other distributions and there are no other distributions to support the aberrant distribution – the distribution is marked "rejected - Mode".

3.2.8 An ECOFF is only determined when there are 5 or more acceptable distributions and at least 5 agree on the mode MIC.

3.2.9 A TECOFF is only determined when there are 3 or 4 acceptable distributions and at least 3 agree on the mode MIC.

3.2.10 Where there are fewer than 3 acceptable distributions in aggregate, the ECOFF will be labelled "ID" – insufficient data to set an ECOFF or TECOFF.

3.3	**MIC distributions acceptable for aggregation (Criteria for validity of aggregation) and for setting ECOFFs** 3.3.1 There must be a single peak in the putative wild-type MIC distribution. Acceptable distributions with a mode that spans two MIC values (i.e. identical counts at adjacent MIC values) in the putative wild type are acceptable. 3.3.2 The range of antimicrobial concentrations tested should be sufficient to include the likely lowest and the likely highest MIC value for the putative wild-type distribution. In order to meet this requirement at least two concentrations on either side of the mode of the putative wild-type distribution must be included in the concentration range. 3.3.3 There must be MIC values for at least 15 isolates in the putative wild-type distribution. 3.3.4 To set an ECOFF, the aggregated MIC distribution must include at least five valid MIC distributions, each of which has been generated in a separate laboratory. This is essential to ensure that inter-laboratory variation is factored into the definition of the reference MIC distribution and for the setting ECOFFs. 3.3.5 The aggregated MIC distribution must contain at least 100 MIC values in the putative wild-type distribution. Although individual distributions must contain at least 15 values, the requirement for a total of at least 100 values means that some distributions must include more than 15 values if only five or six valid distributions are available. 3.3.5 The modes of the individual wild-type distributions in the dataset must be equal to or within one two-fold dilution of the most common mode observed in the collection of valid distributions prior to aggregation.
3.4	**Other datasets** An aggregated distribution that is from only one or two datasets will normally not be displayed on the website. However, for uncommon species, the distribution may be displayed, despite there being no ECOFF set.
3.5	**Other considerations** 3.5.1 Adding a substance that reduces MICs for resistant isolates but has little or no intrinsic antimicrobial activity (e.g. beta-lactamase inhibitors added to beta-lactam agents) does not change the established ECOFF of the original agent. On the other hand, synergistic combinations in which one agent synergistically potentiates the activity of the other or where both components exhibit antimicrobial activity independently, such as sulfamethoxazole added to trimethoprim, merit their own ECOFFs.

	3.5.2 If the putative wild-type distribution after aggregation appears to be more than five dilution steps wide, special attention will be given to possible methodological problems among individual distributions in the dataset. In the experience of the curators the most common reasons for wider distributions are difficulties in reading end-points, systematic methodological variation (such as disagreement on the use of CO_2 incubation) and inaccuracies in species identification. Whether an ECOFF is set will be a consensus decision by the curators of the database (see section 7). 3.5.3 If the aggregated MIC distribution is not symmetrically distributed on either side of the mode (on the log-normal scale), special attention will be given to the possible presence of low-level resistance mechanisms in isolates from one or several of the aggregated distributions in the dataset. Whether an ECOFF is set will be a consensus decision by the curators. 3.5.4 In some instances, phenotypic data other than the MICs of the agent in question or genotypic data (on the presence or absence of known resistance genes) will be available to support ECOFF setting. The collation and storage of such information is beyond the scope of this document but will be addressed in the future.
3.6	**Tentative ECOFFs** 3.6.1 In some cases there is a need for setting a “tentative ECOFF” while collecting data which fulfil the standardised criteria for setting ECOFFs described above. Available data may be scarce because the organism is difficult to work with or a rare species or no standardised method is available or because new data will be slow in coming. This is true for several rare or difficult-to-culture bacterial species and several fungal species. 3.6.2 ECOFFs which are tentative (TECOFF) will be clearly marked as such by displaying them in brackets “(value)”. 3.6.3 Tentative ECOFFs will be reviewed when there are new data. 3.6.4 Tentative ECOFFs will be set for situations where there are three or four acceptable distributions in the aggregate

4 Method used to set ECOFFs

4.1 Statistical + Visual ECOFF estimation

Two complementary approaches to setting ECOFFs will be used: the statistical ECOFF and the visual ECOFF.

The Statistical ECOFF

Each individual acceptable MIC distribution that contributes to the aggregate will be run through the 'ECOFFinder' algorithm which uses an iterative statistical curve-fitting method based on the cumulative log-normal distribution (Turnidge J, Kahlmeter G, Kronvall G. *Clin Microbiol Infect* 2006; 12:418-25) to estimate 99.9% ECOFF of each contributing distribution (log_2 value).

The means and standard deviations of contributing 99.9% ECOFFs are calculated (using their log_2 values). The ECOFF (and the (T)COFF) is the log_2 mean after conversion to its numerical value (2^log_2 mean) and rounding up to the next two-dilution. The log_2 standard deviation is used to determine the 95% confidence interval (on the log_2 scale), after which the lower and upper values of the interval are converted to their numerical vales and rounded down (lower value) and up (upper value) to the next two-fold dilution.

The Visual ECOFF

The visual ECOFF was introduced initially in 2001 when EUCAST started to systematically collect MIC distributions and the first software to store the MIC distributions was created (Kahlmeter et al, *J Antimicrob Chemother* 2003; 52:145-48). The visual ECOFF still plays an important role since 99.9 % statistical and visual ECOFFs are matched as described below:

a) for putative wild-type distributions encompassing five dilutions, the first and last bars of the histogram usually contain 1-5% of the total across those bars

b) for distributions encompassing 3-5 dilutions, the first and last bars usually contain 5-10% of the total across those bars

c) when percentages higher than either of these are observed, the individual contributing distributions prior to aggregation are examined for inconsistent mode and/or width compared to the other contributors; if found they are removed and the rest of the distributions re-aggregated prior to analysis

d) for putative wild-type distributions encompassing more than 5 dilutions, problems related to testing or strain collection in the contributing distributions are sought

    i) methodological problems leading to excessive variability between investigators because of pronounced influence of variations in testing conditions for the species-agent in question

	ii) non-wild type isolates affecting the data sets (significant overlap between wild-type and non-wild type), making the identification of the wild type difficult; this can be rectified by the identification of distributions generated prior to the emergence of non-wild type strains should be sought or by using a more sensitive phenotypic method to distinguish between data representing wild type and non-wild type isolates (e.g. pefloxacin instead of ciprofloxacin in *Salmonella* spp; cefoxitin instead of another beta-lactam agent in *S. aureus*).
4.2	When the statistical and visual approaches agree, the ECOFF is considered to be set. When there is a discrepancy between the visual and the statistical approaches and the discrepancy is no more than one two-fold dilution, the ECOFF will be a consensus decision by the curators. If the discrepancy is more than one dilution, an ECOFF will not be set and more MIC distributions will be sought to strengthen the dataset.

5	**Conditions for revision of ECOFFs**
5.1	As new MIC distributions are added to the database, each will be examined for acceptability as described in section 3 before adding it to the aggregated distribution in the dataset.
5.2	If aggregated distributions already have a published ECOFF, the ECOFF will be re-evaluated (as described in section 4) only if visual inspection indicates that the new distribution(s) may alter the ECOFF. If it is decided that a change in the ECOFF is unnecessary, the new data set is aggregated with the existing data and no further action taken. Should a change in the ECOFF be considered necessary, an alert will be posted on the EUCAST website (see section 6.2) and a list of revised ECOFFs will be provided on the EUCAST website.
5.3	If aggregated distributions previously had insufficient numbers of acceptable distributions in the dataset, these will be determined when sufficient numbers become available (see sections 3 and 4).
5.4	The definitions of a wild-type MIC distribution and the ECOFF are based on phenotypic methods. However, the Subcommittee will examine the influence of known resistance genes on putative wild type distributions. In some cases, the effect of the resistance gene is not detectable by one agent but readily so by another in the same class. For example, the ampicillin MIC distribution for *H. influenzae* may not properly disclose relevant PBP3 mutations, whereas the closely related benzylpenicillin will; or the ciprofloxacin MIC distribution for *Salmonella* spp. may not disclose some fluoroquinolone resistance mechanisms but the use of the closely related pefloxacin will.
5.5	The definition of a wild-type MIC distribution and the ECOFF is normally not affected by the addition of an enzyme inhibitor (amoxicillin vs. amoxicillin-

	clavulanic acid, piperacillin vs. piperacillin-tazobactam, ceftazidime vs. ceftazidime-avibactam etc). The wild type and ECOFF are determined for the parent compound unless the inhibitor has intrinsic phenotypically measurable activity against the species or has been shown to potentiate the action of the parent compound.

6	**Curators of the MIC distribution database**
6.1	The curators of the MIC distribution database will make decisions on the inclusion or exclusion of MIC distributions and on setting ECOFFs.
6.2	The curators of the database are appointed by the EUCAST Steering Committee and listed on the EUCAST website: http://www.eucast.org/mic_distributions_and_ecoffs.

7	**Submission of MIC distributions**
7.1	MIC distributions for any species-antimicrobial agent combination may be submitted for inclusion in the database at any time. The data should meet the standards defined in section 4.2.
7.2	Excel spreadsheets for submission of data are available on the EUCAST website at http://www.eucast.org/mic_distributions_and_ecoffs/. There are two versions of the spreadsheet: • MIC distributions organised by agent or species • Zone diameter distributions organised by agent or species Irrespective of which spreadsheet is used, column "U" (Comments) must be used to include information (* indicates required information) on: • The method* (e.g. ISO broth microdilution, agar dilution on MH, gradient diffusion tests on MH, etc.) • The source of the isolates (human clinical, veterinary, food, etc.) • The name of the study or testing laboratory • The year of the study • The range of dilutions tested* (e.g. 0.06-128 mg/L) • Semicolons (;) and/or commas (,) inside the Excel cells are problematic in data handling. Therefore, the symbol "/" should be used as delimiter between items in the comment column, e.g. "ISO broth microdilution / respiratory isolates / European surveillance study / 2016 / tested range 0.06-128 mg/L". • When possible, MIC distributions included in the spreadsheet should have separate lines for: 1. MICs determined in different laboratories 2. MICs determined by different methods 3. MICs determined in different years of an ongoing surveillance programme

	4. MICs for isolates with known resistance mechanisms (e.g. MRSA or MSSA; extended-spectrum beta-lactamase (ESBL)-producing or non-producing Enterobacteriaceae 5. MICs for isolates of different geographical origin The completed Excel file should be sent to one of the curators (see EUCAST website at http://www.eucast.org/mic_distributions_and_ecoffs/).

Appendix 1. Current introduction to the EUCAST website for MIC and inhibition zone diameter distributions and ECOFFs. (As currently [2021-09-27] at www.mic.eucast.org).

MIC and Inhibition zone diameter distributions of microorganisms without and with phenotypically evident resistance mechanisms

MIC and inhibition zone diameter distributions

Distributions are shown as "aggregated distributions" and as "aggregated weighted distributions". For aggregated distributions all accepted distributions (as defined in SOP 10) were added to form one common distribution. For aggregated weighted distributions each individual distribution was converted to contribute equally to the common aggregated distribution. In this way large distributions are prevented from drowning out smaller distributions.

1. MIC distributions

The website gives MIC distributions for individual micro-organisms (bacteria and fungi) and antimicrobial agents in tables and histograms. The distributions are based on collated data from an increasing total of more than 30 000 MIC distributions from worldwide sources. Unless otherwise specifically stated, the data are representative of results obtained with MIC methods performed by or calibrated to reference broth microdilution using ISO-20776-2. Different methods do not give exactly the same results, but the results rarely vary by more than one doubling dilution step. In this way the aggregated MIC distributions encompass the variation between different investigators, laboratories, geographic locations and time periods.

2. Inhibition zone diameter distributions

The website gives inhibition zone diameter distributions for individual organisms and antimicrobial agents in tables and histograms. The distributions are based on collated data from an increasing number of sources worldwide. The data are representative of results obtained with the EUCAST disk diffusion method (launched in 2009 - see www.eucast.org).

Clinical MIC and Zone diameter breakpoints

These are not shown on this website – please consult the EUCAST breakpoint tables for bacteria and fungi on www.eucast.org.

Epidemiological cut-off values (ECOFF) and tentative epidemiological cut-off values (TECOFF)

ECOFFs (and TECOFFs) distinguish microorganisms without (wild type) and with phenotypically detectable acquired resistance mechanisms (non-wild type) to the agent in question. The epidemiological cut-off value is shown in the tables and the bottom left-hand corner of each MIC and zone diameter graph. TECOFFs (ECOFFs in

parentheses) are based on 3 or 4 distributions and ECOFFs on at least 5 and up to 100 or more distributions.

Limitations

The database is being curated in accordance with EUCAST SOP 10.
Different studies often use different MIC concentration ranges.
Distributions truncated at the lower end of the scale within the putative wild-type distribution have been excluded. The accepted distributions include studies from a wide variety of sources and time periods and some, on purpose, include high and others low, proportions of resistant organisms, so the distributions must not be used to represent rates of resistance to any agent and cannot be used to compare resistance rates among agents, resistance over time or resistance in different geographic locations.

To cite data from the EUCAST program

We suggest the format "European Committee on Antimicrobial Susceptibility Testing. Data from the EUCAST MIC distribution website, last accessed Day Month Year". http://www.eucast.org"

To contribute MIC- or zone diameter distribution data

Please send data to gunnar.kahlmeter@eucast.org or Jenny.ahman@eucast.org. We propose you download from the EUCAST website (https://www.eucast.org/mic_distributions_and_ecoffs/) and use one of the Excel files for bacteria or fungi and follow the instructions in the spreadsheet.

Frequently asked questions

What is the purpose of the MIC distributions?

- reference material for committees involved in decisions on clinical breakpoints
- reference material for epidemiological cut-off values for antimicrobial resistance surveillance
- reference MIC ranges of wild type organisms for a wide spectrum of species and antimicrobials
- international reference for calibration of antimicrobial susceptibility testing methods. If your distribution indicates a different wild type and ECOFF value than that in the database, as evidenced by a putative wild-type modal 2 or more two-fold dilutions different from the mode of the website's aggregated distributions, you probably made a mistake, used inadequate material or have used a method that is neither the reference method or one calibrated to it
- ECOFFs can be used for antimicrobial resistance surveillance purposes, especially when other breakpoints are poorly defined or change over time

What is the source of the data?

The distributions include data from national and international studies and resistance surveillance programs, as well as MIC distributions from published articles, the pharmaceutical industry, veterinary programs and individual laboratories.

Which MIC determination and disk diffusion methods are represented?

The MIC distributions represent MIC values primarily determined with methods described by EUCAST and CLSI using ISO 20776-1 (ISO 16256 for fungi). Other MIC-determination methods clearly calibrated to any of the reference methods using ISO-20776-2 are also included. Where no internationally agreed reference method exists, the distributions are labelled according to the method used.

The zone diameter distributions are based on inhibition zones generated through the EUCAST disk diffusion method only.

Have data obtained with a specific method been excluded?

For MIC data, no specific method has been excluded. Contributions are screened by the EUCAST Subcommittee on Wild Type distributions and ECOFFs and typically 10 - 20% have been excluded from the aggregated distributions, the most common reason being "lower end truncation". However, all data are held in the database and are accessible for discussion.

For inhibition zone diameter data, only data generated with quality-controlled EUCAST disk diffusion methodology have been included. Thus, the upper part, representing organisms without acquired resistance, can be used to calibrate methodology. The wild-type distribution generated in individual laboratories should match the wild type distributions on the EUCAST website (mode within one two-fold dilution).

What is the origin of the organisms included in the data?

The data are from bacteria and fungi collected from man and animals, of any geographic origin and over a period of 70 years.

What does "No Data available" imply?

For some of the antimicrobials in the pull-down list no data is available for public view ("No Data available"). The process of harmonizing clinical breakpoints across Europe and of defining the epidemiological cut-off values is co-ordinated with the collection of relevant MIC data. As decisions on breakpoints are made and the quality of data have been assessed for each group of antimicrobials the data are released for public use.

Where can I get more information?

Contact EUCAST via the EUCAST Contact form. In addition, email addresses and further information can be obtained through the EUCAST website at https://www.eucast.org.

Appendix 2. Gentamicin MIC distribution table (top) and *E. coli* gentamicin MIC graph (bottom) from the EUCAST website (www.mic.eucast.org/).

Search database

Method MIC Disk diffusion

Antimicrobial Gentamicin

Species Species...

Elements per page 50

MIC distributions for Gentamicin, 2021-08-25

Antimicrobial: Gentamicin (Method: MIC)

	0.002	0.004	0.008	0.016	0.03	0.06	0.125	0.25	0.5	1	2	4	8	16	32	64	128	256	512	Distributions	Observations	(T)ECOFF	Confidence interval
Acinetobacter anitratus	0	0	0	0	0	0	0	6	29	37	25	7	5	7	0	0	0	0	0	1	116	ID	
Acinetobacter baumannii	0	0	0	0	0	6	21	226	607	451	150	79	48	80	38	73	36	107	7	22	1929	2	1 - 4
Acinetobacter calcoaceticus	0	0	0	0	0	0	0	6	21	30	18	15	13	7	0	0	0	0	0	1	110	ID	
Actinobaculum schaalii	0	0	0	0	0	0	1	6	9	22	10	0	0	0	0	0	0	0	0	1	48	ID	
Alcaligenes faecalis	0	0	0	0	0	0	0	0	0	1	20	2	0	0	0	0	0	0	0	1	23	ID	
Bifidobacterium longum	0	0	0	0	0	0	0	0	0	0	0	2	8	15	21	9	5	1	1	1	62	ID	
Campylobacter coli	0	0	0	0	4	38	160	1550	4160	1911	175	4	2	1	5	99	0	0	0	50	8109	1	0.5 - 2
Campylobacter jejuni	0	0	0	2	33	141	1688	6829	14541	2346	108	12	5	0	5	62	0	0	1	41	25773	1	0.5 - 2
Citrobacter freundii	0	0	0	1	0	0	17	269	798	363	56	36	37	93	15	55	1	0	32	7	1773	2	1 - 4
Citrobacter koseri	0	0	0	0	0	2	8	42	145	58	12	2	3	1	1	0	1	0	0	3	275	(2)	0.5 - 4
Enterobacter agglomerans	0	0	0	0	0	1	5	43	5	0	0	0	0	0	0	0	0	0	0	1	54	ID	
Enterobacter cloacae	1	0	1	3	26	125	324	1202	3647	1384	160	87	124	92	19	103	50	52	60	38	7460	2	0.25 - 2
Enterococcus faecalis	0	0	0	0	0	2	2	7	18	37	73	156	880	1761	704	61	57	69	686	25	4513	64	32 - 128
Enterococcus faecium	0	0	0	0	0	0	0	1	5	14	83	555	1147	575	93	13	11	46	320	24	2863	32	16 - 64
Enterococcus hirae	0	0	0	0	0	0	0	0	1	4	50	98	234	151	28	0	0	0	2	8	568	32	4 - 32
	0.002	0.004	0.008	0.016	0.03	0.06	0.125	0.25	0.5	1	2	4	8	16	32	64	128	256	512	Distributions	Observations	(T)ECOFF	Confidence interval
Escherichia coli	0	0	4	6	26	49	525	8521	32063	21068	4423	1213	1399	3728	6362	512	138	167	70	93	80274	2	1 - 2
Escherichia coli ATCC 25922	0	0	0	0	0	0	0	1	63	34	7	0	0	0	0	0	0	0	0	2	105	-	
Escherichia coli NCTC 13846	0	0	0	0	0	0	0	0	54	58	8	0	0	0	0	0	0	0	0	2	120	-	

